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CRIMINAL INVESTIGATION IN THE DIGITAL ERA: LESSONS FOR BANGLADESH FROM COLOMBIA, INDIA, AND THE UNITED STATES

INVESTIGAÇÃO CRIMINAL NA ERA DIGITAL: LIÇÕES PARA BANGLADESH A PARTIR DA COLÔMBIA, ÍNDIA E ESTADOS UNIDOS

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ABSTRACT

This paper is a scientific and research-oriented inquiry that examines the interface between technological advances and changes in criminal investigation patterns, specifically identifying lessons that can be used to improve investigative effectiveness, evidentiary integrity, and agency cooperation. The scientific problem at hand is the inability of certain jurisdictions, including the focus nation, to properly handle cyber-enabled crimes and digital evidence and align them with international standards; this deficiency has serious implications for law enforcement effectiveness as well as the protection of rights accorded under due process. The central argument is that learning from a range of operational, legal, and institutional techniques in other legal systems may greatly improve investigative capability in technologically advanced but resource-constrained environments. The objective is to conduct a comparative legal-criminological investigation to identify transferable best practices and provide suggestions for a comparative legal analysis-based reform pattern tailored to the socio-legal context under consideration. The study examines investigation arrangements, digital forensic capacities, and coordination mechanisms using a comparative doctrinal approach, supplemented by content analysis of law texts, institution-based reports, and prominent cases. The findings are unique in that they combine lessons from a variety of legal systems to create a hybrid framework for making recommendations, as well as the scope of legislative provision, institution-based arrangements, and on-the-ground practice in criminal investigations in the digital era. The work provides cognitive value by providing a realistic, evidence-based plan for strengthening investigative ability, which has ramifications for policymakers, criminology scholars, and international law enforcement collaboration.

Keywords: Comparative analysis, Evidence standards, Forensic science Bangladesh, Investigative reforms, Law enforcement technology, Offence-based tools.

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RESUMO

Este artigo é uma investigação científica e orientada à pesquisa que examina a interface entre os avanços tecnológicos e as mudanças nos padrões de investigação criminal, identificando especificamente lições que podem ser usadas para melhorar a eficácia investigativa, a integridade das provas e a cooperação entre agências. O problema científico em questão é a incapacidade de certas jurisdições, incluindo a nação em foco, de lidar adequadamente com crimes facilitados por meios digitais e provas digitais, alinhando-as aos padrões internacionais; essa deficiência tem sérias implicações para a efetividade da aplicação da lei, bem como para a proteção dos direitos assegurados pelo devido processo legal. O argumento central é que aprender com uma variedade de técnicas operacionais, jurídicas e institucionais de outros sistemas legais pode melhorar significativamente a capacidade investigativa em ambientes tecnologicamente avançados, mas com recursos limitados. O objetivo é realizar uma investigação jurídico-criminológica comparativa para identificar boas práticas transferíveis e apresentar sugestões para um modelo de reforma baseado em análise jurídica comparada, adaptado ao contexto sociológico e jurídico em questão. O estudo examina arranjos investigativos, capacidades de perícia digital e mecanismos de coordenação por meio de uma abordagem doutrinária comparativa, complementada por análise de conteúdo de textos legais, relatórios institucionais e casos proeminentes. Os achados são inéditos ao combinar lições de diversos sistemas legais para criar um quadro híbrido de recomendações, abrangendo o escopo das disposições legislativas, arranjos institucionais e práticas operacionais nas investigações criminais na era digital. O trabalho oferece valor cognitivo ao apresentar um plano realista e baseado em evidências para fortalecer a capacidade investigativa, com implicações para formuladores de políticas públicas, pesquisadores da criminologia e iniciativas de cooperação internacional em aplicação da lei.

Palavras-chave: Análise comparativa, Padrões de prova, Ciência forense Bangladesh, Reformas investigativas, Tecnologia para aplicação da lei, Ferramentas baseadas em tipificação de delitos.

Introduction

Background and Problem Statement

The globally criminal behaviour landscape has changed dramatically over the last decade, owing mostly to the quick speed of technical improvement and greater usage of digital media. The prevalence and severity of cybercrime, as well as technology-supported crimes, have expanded dramatically, presenting itself in a variety of ways such as identity theft, ransom attacks, financial crimes organised through the virtual world, cyber terrorism, and cyberespionage. The global rise in cybercrime has been a source of worry for law enforcement, necessitating a significant expansion in investigation techniques and legislative frameworks. According to UNODC estimates, cybercrime affects millions of individuals worldwide each year, with financial losses totalling more than $6 trillion (UNODC,

2022). The virtual world not only allows for the continuation of age-old crimes through contemporary methods, but it also gives rise to a slew of new unlawful activities that transcend territorial boundaries and legal authorities (Holt & Bossler, 2022).

Simultaneously, the area of criminal investigation is experiencing a huge upheaval as a result of digital technology breakthroughs. Police forces around the world have recognised the need to meet these new challenges by incorporating sophisticated technologies such as digital forensics, machine learning, large-scale big data analysis, and blockchain verification for evidence corroboration (Mokhtarian et al., 2023). The capacity to collect, maintain, and analyse electronic evidence, which is an essential component of the digital forensic toolkit, has become important to detecting and prosecuting cybercrimes (Choo & Smith, 2022). At the same time, international collaboration approaches, assisted by international organisations like as INTERPOL and national-level cybercrime task forces, have been implemented to address the uniquely transnational aspect of cybercrime (Levi, Burrows, & Holt, 2023). García & Restrepo (2022) argue that traditional policing institutions are unable to address the complex and ever-changing risks posed by cybercrime organisations, highlighting the need for digital augmentation of investigative tactics globally.

The ramifications of this worldwide change are particularly serious for Bangladesh, which is already undergoing massive digitalisation in government, industry, and society. Bangladesh's digital environment is fast evolving, with more than 100 million people obtaining internet access and an increasing trend towards smartphone use, while also becoming more vulnerable to cyber-attacks (Bangladesh Telecommunications Regulatory Commission [BTRC], 2024). The number of cybercrime incidents in Bangladesh has steadily grown, ranging from financial theft and data leakage to cyberbullying and increasingly sophisticated attacks on vital infrastructure (Ahmed & Rana, 2023). Despite the implementation of legal measures such as the Digital Security Act 2018 and the recently approved Cybersecurity Ordinance 2025, Bangladesh's criminal justice system faces substantial challenges in detecting and prosecuting technology-facilitated crime. These include insufficient

institutional capacity, a scarcity of competent specialists, outdated forensic techniques, and hurdles to inter-agency collaboration (Chowdhury, 2024; Hossain & Karim, 2022).

Bangladesh requires action to increase its digital investigation capabilities, motivated by the socioeconomic ramifications of cybercrime, including its impact on national security, public confidence, and digital economy growth. Furthermore, Bangladesh's geographical position, along with escalating cyber dangers, emphasises the need for a unified investigative approach that answers to national demands while being consistent with international norms and attentive to regional dynamics (Rahman, 2023). As a result, a critical review of sophisticated digital investigative systems from other areas with comparable or greater capabilities can be beneficial to Bangladesh's policymaking and strategic planning efforts.

Literature Review

The academic and policy-based literature on digital criminal investigations has increased significantly over the last decade, following the evolution of cyber risks. Wall (2023) provides a thorough overview of the dynamics of transnational cybercrime and the evolution of investigative tactics, emphasising digital forensics and intelligence-led policing. In a separate research, Holt and Bossler (2022) examine reactions to transnational cybercrime, focussing on the variety of organisational structures and technology utilised in various countries.

The past research focussing on Colombia highlight DIJIN's pioneering role in integrating intelligence operations with digital forensic capabilities to address the problem of cybercrime. According to García and Restrepo (2022), Colombia's Criminal Investigation Directorate established proactive cyber teams and collaborated with INTERPOL to effectively combat foreign cybercriminals. González and Martínez (2023) provide insights on Colombia's Tactical Cybercrime and Intelligence Laboratory (TC2IL), which enables real-time intelligence fusion and swift response measures.

The Indian experience provides vital insights for addressing cybercrime in the face of significant socioeconomic disparities. Patel and Verma (2024) describe the Indian Computer Emergency Response Team (CERT-In), which is a centralised entity responsible for coordination and is backed by a network of digital forensic laboratories across regions. Patel and Verma's work show India's cost-effective and creative methods to capacity building, such as decentralised training with an emphasis on maximum local responsiveness (Choo and Smith, 2022). Furthermore, legislative advances, including modifications to the Information Technology Act, has bolstered India's legal structure for digital investigations; nonetheless, obstacles during execution persist (Singh & Rajan, 2023).

The cyber investigation landscape in the United States is complicated, with powerful legal structures and specialised authorities. Kerr (2022) and Solove (2023) examine significant Supreme Court judgements in the United States that affect the admission of digital evidence and the preservation of privacy rights. Jones and Hall (2022) also investigate the complicated federal-state interactions exemplified by the FBI Cyber Division's engagement in national cybercrime investigative strategy. The US legal system prioritises crucial characteristics of effective cybercrime law enforcement, including as high procedural accuracy, public-private sector collaboration, and ongoing technology improvements (Levi et al., 2023; Furnell & Clarke, 2022).

Despite increased academic interest, Bangladeshi literature on digital criminal investigations is still limited and primarily descriptive. Chowdhury (2024) critiques the Digital Security Act, pointing up its ambiguous clauses and potential conflicts with free expression. At the same time, Hossain and Karim (2022) find serious gaps in the institution's capacity on the side of both the judiciary and the police. Rahman (2023) advises against large changes to Bangladesh's cyber legal and investigative infrastructure, emphasising the importance of standardisation with international standards and interagency collaboration. Ahmed and Rana (2023) emphasise the vital need for talent development and infrastructure enhancements, and acknowledge the absence of empirical research on real investigation methods from Bangladesh.

This study is significant since there have been few extensive and comparative investigations of the digital investigation landscape in Bangladesh. The research, which examines investigative frameworks implemented in the United States, India, and Colombia, seeks to identify acceptable best practices and give recommendations based on evidence tailored to Bangladesh's specific sociolegal setting.

Purpose and Objectives of the Study

Given the enormous challenges posed by cybercrime and the evolving international approaches used to combat these threats, this study seeks to investigate the methods used in cybercrime investigations in selected jurisdictions, namely the United States, India, and Colombia, and to provide findings that can be used to improve investigative competence in Bangladesh.

The primary research topic being studied is: How can Bangladesh successfully design and deploy digital inquiry models to combat technology-enabled criminal behaviour, drawing on experience from Colombia, India, and the United States?

To respond to this question, the research is structured around the following questions:

i. What are the key institutional, legal, and operational characteristics of Colombia, India, and the United States' digital criminal investigative systems?
ii. How do these frameworks operate digital forensics difficulties, interagency cooperation, and the legal admissibility of digital evidence?
iii. What best practices from these jurisdictions may be applied to boost Bangladesh's digital investigative capabilities?

The research aims to fulfil the following particular objectives:

i. Conduct a thorough comparative examination of the procedures and approaches used in the three jurisdictions mentioned.
ii. Identify and adapt successful methodologies and paradigms for the Bangladesh setting.
iii. Recommend policy measures to improve Bangladesh's digital investigative system, as well as legal and institutional capacities.

With achieving these objectives, the research aims to contribute to both the intellectual discussion of comparative criminology and the development of pragmatic policy within the Bangladesh context. The future findings are supplemented to enable the development of a more effective, efficient, and human rights-compliant criminal justice system response to cybercrime, while increasing public trust and strengthening national security in the information age.

Methodology, methods and materials

Type of Study and Research Design

This study is based mostly on a qualitative and doctrinal research technique. The paper uses a comparative legal-criminological method to critically examine the regulatory frameworks for digital criminal investigations in Bangladesh and three comparator jurisdictions: the United States, India, and Colombia. The doctrinal form of research entails a thorough review and analysis of legal literature, organisational procedures, and important case law pertaining to cybercrime investigations. The technique seeks to create a greater awareness of substantive and procedural legal requirements essential to investigative work. A qualitative analysis of policy is also included, providing a critical reflection on the organisational structures, strategies, and strategic planning used by law enforcement institutions in response to the complex challenges posed by digital forensics and cyber policing. Individually, these methods provide a comprehensive framework that combines the normative legal environment with the practical consequences of digital criminal investigations.

Data Sources

The data provided here is a combination of primary and secondary sources. Statutory law, criminal procedural law, cybercrime legislation, and laws governing the acquisition and admissibility of digital evidence are among the primary sources used here. Institutional sources, such as official law enforcement agency records, digital forensic techniques, and national policy announcements, also provide useful information. Aside from this, court decisions and case law that codify these legal frameworks provide support for the doctrinal approach. Secondary sources include peer-reviewed scientific publications, government and institutional reports, and journals from international agencies such as INTERPOL and the United Nations Office on Drugs and Crime (UNODC). Secondary sources include relevant analytical commentary, empirical data, and historical knowledge that help understand original documents and give comparative analysis across different legal jurisdictions.

Comparative Jurisdiction Selection Criteria

Colombia, India, and the United States were chosen as comparator nations because to their unique relevance to Bangladesh's digital investigative prospects and difficulties. Colombia serves as a model case study due to its pioneering policing policies, particularly through the Criminal Investigation Directorate (DIJIN), as well as its collaboration with INTERPOL to combat transnational cybercrime. India's decision is justified by its socioeconomic and developmental parallels with Bangladesh, including shared issues such as infrastructural deficiencies and the establishment of digital infrastructures. India's efforts to establish centralised procedures for public reporting of cybercrime, as well as to develop specialised forensic training infrastructures, provide reproducible and applicable examples for Bangladesh. The United States has a well-established legal and operational framework for digital forensics, which includes broad statutory laws, specialised cybercrime units, and effective integration of technology capabilities. The chosen nations present a diverse variety of approaches for identifying prospective 'best

practices' and benchmarking them against Bangladesh's legislative and institutional context.

Instruments and Apparatus

Research use specialised qualitative methodologies to conduct a systematic study of collected data. Content methodologies are used to analyse policy documents and organisational policies, allowing for the identification and classification of investigation procedures, information system integrations, inter-agency cooperation, and capacity building initiatives. Statutory analysis tools are used to conduct comparative analyses of legislative legislation, with an emphasis on elements such as scope, definitions, procedural mandates, and evidentiary standards used in digital investigations. The analytical tools enable a complete and systemic investigation of legislation and organisational structures, ensuring comprehensive and accurate comparisons across countries.

Procedure

The research approach employed a methodical approach to gather and compile pertinent legal materials, case law, institution policy, and secondary sources from reliable sources such as government websites, international organisations, and legal databases. Following data collection, qualitative theme coding was applied, and data that satisfied the study's objectives was extracted using established coding structures. Principal topics pertaining to legislative infrastructure, everyday difficulties, digital forensic capability, and international collaboration were identified during the coding process. Following consolidation of the collected data to create comprehensive jurisdictional profiles, comparative analysis was performed to identify key best practices, similarities, and differences. The examination of legal sources in this study was guided by doctrinal analysis, while the practice of the institution and reform initiatives were framed by qualitative policy analysis.

Ethical Consideration

There is no direct interaction with human subjects or animals in this research, and there are no significant ethical concerns because it mostly depends on existing legal documents, institutional regulations, and academic literature. By correctly citing all sources and avoiding intellectual property violations, the research closely complies with ethical standards. Clarity and respect for research are enhanced by the rigorous regulations around citation and referring that follow the guidelines of the APA 7th edition. Furthermore, the study methodology has been meticulously documented to guarantee both academic integrity and replicability.

Results

Jurisdictional Findings

Colombia, India, and the United States have distinct but complementary systems for countering cybercrime, including advanced digital forensic capabilities, legislative frameworks, institutional administration, and international collaboration. These jurisdictions give a deeper understanding of how governments might use technology in criminal investigations to successfully control modern cyber dangers.

Colombia

In Colombia, the Directorate of Judicial Investigation and Intelligence (DIJIN) is the centrepiece of cybercrime investigations, with a strong emphasis on digital forensics and intelligence-based policing. DIJIN has specialised cybercrime divisions to combat sophisticated cybercrimes against financial institutions, infrastructure, and individuals (García & Restrepo, 2022). The centres use cutting-edge digital forensic labs capable of extracting, storing, and processing electronic evidence from a variety of sources such as PCs, mobile devices, and cloud settings. DIJIN's forensic capability is enhanced by ongoing staff training to keep investigators up-to-date with new technologies and procedures (González & Martínez, 2023). DIJIN's affiliation with INTERPOL and other international law enforcement organisations is

one of the most notable aspects of Colombia's cyber investigative capabilities. This network enables the real-time exchange of intelligence, coordinated multinational operations, and the prosecution of transnational crimes. Joint measures against ransomware cartels and cybercrime networks operating online, for example, have proved the value of such coordination (Holt & Bossler, 2022). DIJIN may also gain access to global cyber threat intelligence through these relationships, which improves its ability to foresee developing cybercriminal techniques.

In addition, the UK has enhanced its intelligence-led police approaches, in which cybercrime teams collaborate with analysts who use big data and behavioural analytics to find patterns and anticipate criminal activity. The Tactical Cybercrime and Intelligence Laboratory (TC2IL) is an example of such an initiative, combining open-source intelligence, social media tracking, and forensic data to provide actionable intelligence products (Wall, 2023). Despite this, Colombia has challenges such as regional disparities in forensic capabilities and the need to constantly adjust legislative laws to confront emerging forms of cybercrime, notably cryptic currency and the dark web (Mokhtarian et al., 2023).

India

In India, digital investigative strategies will take into account the country's predicted demographic variety, decentralised governance structure, and growing legislation. The Indian Computer Emergency Response Team, or CERT-In, is India's national coordinating agency for reacting to cybersecurity crises, such as cybercrime investigations. The CERT-In publishes technical advisories, shares threat intelligence, and provides capacity-building initiatives for private sector and law enforcement participants (Patel & Verma, 2024). Central coordination allows for a timely reaction by the entire nation to cyber problems, which is especially important in a country undergoing rapid digitalisation and increased cybercrime complaints.

India's legal system is founded on the Information Technology Act of 2000 and its revisions, which legalise cybercrime, allow for electronic evidence

admissibility, and govern intermediary responsibility (Singh & Rajan, 2023). The Indian Evidence Act, which has been interpreted by the courts to include electronic evidence, supplements the Act's requirements. The rules governing digital evidence and forensic analysis are changing, but consistency is lacking owing to federal-state jurisdictional overlaps.

The Indian cyber forensic infrastructure is made up of a network of Ministry of Home Affairs-accredited regional cyber forensic laboratories that allow for the fast investigation of digital evidence across the country (Patel & Verma, 2024). Capacity development is prioritised, with organisations offering investigator training, court sensitisation training, and public outreach training. Notably, India uses decentralised training systems that enable regional police schools and forensic laboratories to provide customised seminars based on local needs (Choo & Smith, 2022). Partnerships with technology companies and educational institutions improve training quality.

Despite improvements, challenges persist. Coordination deficiencies between the federal and state governments are most likely to impede investigations. The digital gap and limited infrastructure in rural regions limit homogenous capability, and the rapid speed of technological change necessitates continuous upskilling (Holt and Bossler, 2022). Furthermore, privacy issues and data protection debates hamper legislative reform, necessitating a delicate balance between security and rights.

United States

The United States is an advanced, highly institutionalised form of digital investigations, with strong federal-state cooperation, specialised authorities, and a well-established judicial system. The FBI's Cyber Division investigates cyber intrusions, ransomware, identity theft, and other computer-related crimes in collaboration with state and local law enforcement and foreign partners (Furnell & Clarke, 2022). The US system is distinguished by multi-layered institutional specialisation, with agencies such as the Department of Homeland Security, Secret

Service, and Cyber Command supplementing each other's experience and intelligence.

The United States has established a mature jurisprudence that governs digital evidence collecting, privacy rights, and due process. Landmark Supreme Court cases like Carpenter v. United States and Riley v. California limited government access to mobile phone data and the necessity for warrants, creating precedents that strike a balance between investigative power and Fourth Amendment safeguards. These verdicts have consequences for procedural standards governing the admission of digital evidence and evidence collection methods across the country.

The Federal Rules of Evidence include clear advice for electronic evidence, and the United States has developed specialised prosecution guidelines and training programs to improve judge expertise in cybercrime cases. The National Cyber Forensics and Training Alliance (NCFTA) is the gold standard of public-private partnerships, carrying out research, delivering threat intelligence, and providing training to law enforcement and industry (Levi et al., 2023).

The United States has persistent challenges, such as jurisdictional overlaps among federal, state, and local entities, as well as the growing complexity of cyber threats. To solve these issues, sustained investment in laboratory modernisation, human resource capacity building, and legal change is necessary (Jones & Hall, 2022). The US approach prioritises flexibility, broad legal protection, and coordination among different stakeholders in order to maintain robust cyber investigative capabilities.

Bangladesh's Current Framework

In comparison to the countries mentioned above, Bangladesh's digital investigative system is still in its infancy, despite tremendous development over the past 10 years. Significant operational restrictions, newly formed specialised entities, and severe legislative innovation characterise Bangladesh's institutional and legal environment for cybercrime management.

Legal Instruments

The primary law addressing cybercrime in Bangladesh until 2025 was the Digital Security Act (DSA) of 2018. It made a number of crimes illegal, such as data theft, cyberterrorism, online defamation, and illegal access (Hossain & Karim, 2022). The Act also made provisions for electronic evidence and guidelines for cybercrime investigation and prosecution procedures. Notwithstanding its extensive reach, the DSA had drawn criticism for its vague wording, the considerable discretionary authority granted to law enforcement, and possible violations of free speech (Chowdhury, 2024). Both legal scholars and civil society have expressed worry about the absence of clear procedural protections and court monitoring, which might lead to an intervention to strike a balance between security requirements and human rights.

The Evidence Act of 1872, which also pertains to the admissibility of evidence generally, including electronic data, joins the DSA. However, because it was developed during the colonial era, it lacks the specificity necessary to handle intricate digital evidence, which poses interpretive difficulties for judges.

A significant turning point in the country's legislative history was later reached in 2025 when Bangladesh approved the Cybersecurity Ordinance (Ministry of Law, 2025). By creating inter-agency cooperation mechanisms, simplifying the processes for digital evidence, and improving the definitions of cyber offences, the law significantly enhances the DSA. Additionally, it creates a national cybersecurity framework with the goal of fostering international collaboration and public-private partnerships. Although this ordinance is a good piece of legislation, sufficient institutional support and capacity building are necessary for its successful implementation.

Institutional Capabilities

Operationally, Bangladesh's criminal justice system combats cybercrime largely through the Police Cyber Unit, the Criminal Investigation Department (CID),

and newly formed Cyber Tribunals. The Police Cyber Unit conducts preliminary investigations and gathers digital evidence. CID, with its specialised cybercrime cells, handles more complicated cases, which generally involve cybercriminal syndicates and transnational offences (Ahmed & Rana, 2023). Cyber Tribunals were developed to alleviate judicial backlogs and increase decision quality by appointing judges with expertise in cybercrime and technology (Yesmen & Ahmed, 2022).

However, the institutional foundation is seriously jeopardised. The forensic infrastructure remains poor, with few regional digital forensic labs and equipment that has not been significantly modernised (Mokhtarian et al., 2023). The scarcity of trained cyber forensic professionals puts a strain on human resources, and professional training for law enforcement and the judiciary is inconsistent and restricted. Coordination between police, CID, Cyber Tribunals, and other organisations is generally haphazard, resulting in duplication of efforts and inefficiencies.

Furthermore, court understanding of cyber laws and electronic evidence is insufficient, resulting in procedural blunders and unpredictable outcomes. Investigation delays and punishment exacerbate public discontent with the system's ability to effectively combat cyber risks (Rahman, 2023).

Gaps and Challenges

Bangladesh has significant structural inefficiencies that prevent successful digital investigations. These include financial inefficiencies, such as funding shortages in forensic labs and training courses; technological inefficiencies, such as obsolete software and hardware; and operational inefficiencies caused by a lack of inter-agency collaboration (Yesmen & Ahmed, 2022). The lack of a national-level cybercrime coordination centre, such as India's CERT-In or Colombia's TC2IL, exacerbates these challenges.

Furthermore, the digital gap in Bangladesh hinders access to technology and cyber literacy in rural and underserved regions, lowering the efficacy of reporting systems and preventative education (Mokhtarian, Alston, & Chen, 2023). Public

campaigns against cybercrime are weak, resulting in low rates of reporting and delayed responses. Despite these limitations, Bangladesh has been working to improve cyber investigative capability through international partnerships, especially with INTERPOL and regional associations. These initiatives are promise for training, intelligence exchange, and operational assistance, but they are still in the process of becoming institutionalised inside their own nation.

Comparative Synthesis

The comparative study finds that Colombia, India, and the United States have highly sophisticated, well-funded digital investigative systems with specialised teams, legislative transparency, inter-agency collaboration, and ongoing capacity building. Bangladesh's system, while rapidly evolving as a result of laws and institutional foundation, is nonetheless underdeveloped and plagued by structural, resource, and functional challenges.

Colombia and the US have modern cybercrime laws with precise definitions and limitations on digital evidence admissibility (García & Restrepo, 2022; Kerr, 2022). India's IT Act and revisions are adequate legislative codes; however, they require harmonisation across jurisdictions (Singh & Rajan, 2023). Bangladesh's Cybersecurity Ordinance and Digital Security Act establish a basic legal framework that requires future development to clarify procedural protections and balance rights protection (Ministry of Law, 2025; Hossain and Karim, 2022).

Colombia and the US have centralised cybercrime units with strong international collaboration (González & Martínez, 2023; Furnell & Clarke, 2022). India uses an approach that blends centralised coordination via CERT-In with decentralised regional forensic laboratories and training (Patel & Verma, 2024). Bangladesh's decentralised police and CID cyber squads, as well as Cyber Tribunals, are examples of early-stage institutional specialisation that require improved coordination and capacity investment (Ahmed & Rana, 2023).

Capacity building is an important difference. The United States and India have well-established investigative, prosecutorial, and judicial training systems that

are backed by academic and corporate partnership (Levi et al., 2023; Choo & Smith, 2022). Colombia follows an intelligence-driven strategy in which technical skills complement strategic analysis (Holt & Bossler, 2022). Bangladesh's efforts are in their early stages, with ad hoc training and insufficient judicial education being the most important areas for advancement (Mokhtarian et al., 2023).

International cooperation is vital. Colombia and the US have successfully collaborated with INTERPOL to conduct cross-border cybercrime investigations (González & Martínez, 2023; Jones & Hall, 2022). India is actively involved at the regional level, with opportunities to expand international engagement. Bangladesh's worldwide influence is expanding, but formal institutional frameworks are required to maximise impact (Chowdhury, 2024).

Finally, Bangladesh's cybercrime investigation model may learn from these countries' laws, institutions, and operations, which can then be modified to its socioeconomic and technical setting. Clarity in the legislation, specialist institutions, capacity development, and coordination with global partners may help Bangladesh improve its cybercrime response and administer justice in the digital age.

Discussion

Interpretation of Findings

The comparative analysis of Colombia, India, the United States, and Bangladesh yielded results that are both congruent and divergent with the current literature on digital criminal investigation and its growing legal, institutional, and functional environment. The observed countries' practices and concerns reflect the increased global trend of cybercrime and the need for effective digital investigative methods, as mentioned in the introduction. But. The research also identifies minor differences that affect regional socioeconomic, political, and technical settings.

The study of past literature focused on the growing sophistication of cybercrimes and the critical role that institution-specific structures play in combating them (Wall, 2023; Holt & Bossler, 2022). Our findings support this

notion, notably in Colombia and the United States, where highly specialised cybercrime units such as DIJIN's cyber forensic unit and the FBI Cyber Division play critical roles. García and Restrepo (2022) and Furnell and Clarke (2022) identify jurisdictions that successfully combine operational intelligence, advanced forensic technologies, and worldwide collaboration. Colombia's proactive, intelligence-led policing program, which includes the Tactical Cybercrime and Intelligence Laboratory, is consistent with international best practices reported in recent criminological research (Mokhtarian et al., 2023). The emphasis in operations on proactive detection and response differs from more reactive techniques elsewhere.

India's approach, with its decentralised training infrastructure and central coordinating function of CERT-In, partially supports Patel and Verma's (2024) empirical evidence for the efficacy of capacity-building tailored to local conditions and resource limitations. According to the report, "India's cost-effective innovation strategy and regional forensic laboratories enable greater geographical extension and accessibility" corresponds with Choo and Smith's (2022) focus on adjusting cyber-investigative competence to various contexts.

The jurisprudential clarity and institutional competency of the United States in the admission of digital evidence are consistent with Kerr's (2022) and Solove's (2023) assessments of US legal evolution. Our results recognise how model court judgements shape procedural quality and protect constitutional rights, resulting in a more predictable and transparent inquiry environment. This corroborates the major role that standard legal frameworks have in establishing technological and functional developments in cybercrime investigations (Jones & Hall, 2022).

On the other hand, Bangladesh's fast changing yet relatively fresh legal and institutional structure differs significantly from the models of the other three nations. While the Digital Security Act and Cybersecurity Ordinance are significant legislative accomplishments, implementation gaps and confusing regulations governing the admission of digital evidence provide continued issues, as noted by Chowdhury (2024) and Hossain and Karim (2022). This is consistent with the wider literature, which presents evidence of the problems that poor nations have in realigning customary legal codes with new digital realities (Rahman, 2023).

Bangladesh's narrow institutional specialisation and disconnected coordination contrast sharply with more cohesive and resource-rich frameworks in the cohesive States, Colombia, and India. This heterogeneity lends credence to the hypothesis in global criminology studies that cyber investigative competence is strongly associated with governance maturity, resource capacity, and institutional coherence (Mokhtarian et al., 2023; Levi et al., 2023).

In summary, the findings support the crucial need of a comprehensive approach that includes legislative openness, institutional specialisation, inter-agency collaboration, and ongoing capacity training for the finest digital investigations. They also demonstrate that using worldwide best practices necessitates careful modification to local conditions, such as socioeconomic realities and governance systems. This view supports the point made by Choo and Smith (2022) and Mokhtarian et al. (2023) that cybercrime remedies must be context-dependent in order to be sustained and successful.

Lessons Learned

Comparative jurisdictional analysis provides the following insights that Bangladesh might apply to improve its digital investigation skills. These lessons cover operational intelligence, training and capacity building, legal reform, and inter-agency collaboration.

Lessons from Colombia

Colombia's strength stems in its operational intelligence posture, as well as highly involved cyber divisions. DIJIN's combination of cyber forensics and intelligence analysis enables early detection and response to cyber threats, a widely recognised best practice (García & Restrepo, 2022). The emergence of cybercrime teams capable of coordinating with international partners, like as INTERPOL, emphasises the importance of strong international collaboration procedures. Colombia's successes in collaborative operations against ransomware gangs and

international fraud demonstrate the benefits of intelligence-driven police and operational flexibility (Holt & Bossler, 2022). Bangladesh might benefit from establishing such intelligence fusion centres that combine digital forensic skills with strategic intelligence to detect cyber threats before they become severe.

Colombia invests in ongoing training and forensic technologies to maintain operational relevance in a rapidly changing digital world (González & Martínez, 2023). This proactive approach differs from reactive or patchwork training models, emphasising the significance of ongoing professional growth. Bangladesh's minimal but expanding training activities might be modelled after Colombia's integration of information and training to improve investigator preparedness.

Lessons from India

India demonstrates cost-effective innovation and decentralised capacity-building activities that are particularly well-suited to resource-constrained environments such as Bangladesh. CERT-In's position as a central coordinating organisation for handling cybersecurity events, supported by a chain of regional forensic laboratories, exemplifies the capacity of decentralisation to enhance coverage and responsiveness at an outrageous cost (Patel & Verma, 2024). This distributed system provides virtual access to digital forensic services even in rural areas, solving the geographical variance and digital divide issues that Bangladesh faces (Mokhtarian et al., 2023).

India's emphasis on decentralised training, utilising local police academies and academic partnerships, promotes localised capacity building that is responsive to local needs (Choo and Smith, 2022). This modular professional development strategy facilitates skill scaling while decreasing reliance on a central agency, which Bangladesh's Police Cyber Unit and CID may use to broaden training breadth and customise programs locally.

Legislative flexibility, as demonstrated by the updated version of India's Information Technology Act and judicial interpretation of electronic evidence, provides a viable model for Bangladesh to construct its own legislation. India's

ongoing attempts to balance cybersecurity concerns and privacy rights through public discussions and judicial review processes demonstrate the value of participatory policymaking (Singh & Rajan, 2023).

Lessons from the United States

The United States is a jurisprudential paradigm for strength, institutional specialisation, and inter-agency collaboration in digital investigations. The US Supreme Court's historic decisions provide a strong legal framework that protects constitutional rights while allowing for the acquisition of digital evidence (Kerr, 2022; Solove, 2023). Bangladesh should learn from the United States' emphasis on procedural accuracy and judicial experience when addressing cybercrime matters in order to decrease evidence uncertainty and enhance trial outcomes.

The FBI Cyber Division and coordinated federal-state enforcement highlight the value of explicit role delineation, resource specialisation, and established inter-agency frameworks (Furnell & Clarke, 2022). Bangladesh's present institutional arrangements should be enhanced further by establishing separate national cybercrime investigative organisations with well-defined missions and resources, which would improve cooperation between police, CID, and Cyber Tribunals.

Furthermore, the American approach demonstrates the need of public-private partnerships and threat intelligence sharing channels, such as the National Cyber Forensics and Training Alliance (Levi et al., 2023). These relationships allow for the real-time exchange of threat intelligence, operational assistance, and joint training programs, which Bangladesh may emulate in its cybersecurity practice to harness the private sector's experience and infrastructure.

Policy Implications for Bangladesh

Lessons from comparative countries reveal a number of important policy implications for Bangladesh's digital investigation capabilities development.

Legal Clarity on Digital Evidence Admissibility

A key policy objective is to establish the legal basis for admitting digital evidence. Bangladesh's present regulations, such as the Digital Security Act and Cybersecurity Ordinance, are ambiguous about evidentiary standards, chain of custody, and the scope of permissible digital data acquisition (Hossain & Karim, 2022; Federal Ministry of Law, 2025). Drawing on the US experience, legal reform should include explicit regulations that are compatible with constitutional safeguards, judicial guidance, and global standards (Kerr, 2022; Solove, 2023). This will eliminate procedural issues, boost prosecutorial trust, and improve judicial uniformity.

Strengthening Inter-Agency Coordination and Technology Partnerships

Effective cybercrime investigations need flawless collaboration among the judiciary, law enforcement, intelligence agencies, and private sector actors. Bangladesh's decentralised coordination methods impede timely information exchange and operational efficiency (Ahmed & Rana, 2023). Establishing centralised coordination centres, such as India's CERT-In or Colombia's TC2IL, can improve intelligence fusion, shared operations, and resource optimisation (Patel & Verma, 2024; García & Restrepo, 2022).

Furthermore, formal cooperation with telecommunications companies, internet service providers, and cybersecurity providers can give technological skills and access to threat intelligence (Levi et al., 2023). Facilitating such public-private collaboration will increase investigation depth and pace while using current technical infrastructure.

Emphasis on Investigator Skill Development

Continuous capacity growth is crucial. Bangladesh's poor cyber forensic skills and unbalanced training programs provide substantial challenges to successful investigations (Mokhtarian et al., 2023). Policy intervention must include comprehensive and ongoing training for investigators, prosecutors, and judges on digital forensics, legal frameworks, and developing cyber dangers (Choo & Smith,

2022). Decentralised training methods based on regional workshops and academic collaborations may improve accessibility and flexibility (Patel & Verma, 2024).

Spending in forensic infrastructure modernisation requires skill development to provide staff with contemporary technology and analysis tools (González & Martínez, 2023). Additionally, IT-expert professional-focused recruiting drives can help to establish a long-term investigative force.

Limitations

While this study provides intriguing data, it has a few fundamental shortcomings. For starters, without empirical interviews or surveys of Bangladeshi investigators, court officials, and politicians, our grasp of operational realities, internal constraints, and cultural variables influencing digital investigations is limited. Empirical evidence may include delicate impressions of procedural obstacles, resource distribution, and inter-agency collaboration that cannot be gathered from doctrinal investigation.

Second, possible underreporting or nondisclosure of internal difficulties and operational procedures in Colombia, India, and the United States may distort compared results. Some governments may keep important cyber activities hidden or lack detailed public records of their investigation methods.

Third, the three comparison jurisdictions restrict the extent of generalisability, particularly given the different cybercrime contexts in other locations. Future study might widen the comparative view to include other nations with similar socio-political and economic features to Bangladesh.

Future Research Directions

According to the current study's findings and limitations, future research should prioritise empirical investigations involving key players in Bangladesh's criminal justice and cybersecurity systems. Qualitative interviews, focus groups, and ethnographic studies might shed light on operational issues, institutional culture,

and capacity deficiencies from the viewpoints of frontline investigators, prosecutors, and judges.

Furthermore, quantitative study evaluating the efficacy of digital forensic technologies, case handling time, and prosecution rates might offer numerical estimates of investigative capability.

A comparative analysis at the regional level in South Asia, incorporating Sri Lanka, Nepal, Pakistan, and Bhutan, might help to contextualise Bangladesh's situation and inform regional cybercrime action. This would allow for the exchange of information about shared challenges and the creation of collaborative structures capable of combating common digital dangers.

Furthermore, cross-disciplinary study combining legal studies, criminology, information technology, and public policy will improve knowledge of cybercrime investigation dynamics and help in the creation of comprehensive solutions tailored to Bangladesh's socio-technical setting.

Conclusions

Summary of Objectives and Key Findings

The research aimed to analyse and compare the methods and structures of online criminal investigations in Colombia, India, and the United States, with the goal of drawing relevant lessons for Bangladesh. The primary goals were to examine the legislative, institutional, and operational systems that enable cyber investigations in the selected countries, to identify the most effective practices that can be tailored to the Bangladeshi context, and to propose reforms to strengthen Bangladesh's cyber investigative capacity. The research used a qualitative and doctrinal comparative technique to incorporate legal texts, institutional guidelines, and academic literature, resulting in an overall assessment of each jurisdiction's strengths and shortcomings.

The study fulfilled its objective by outlining Colombia's externally linked, intelligence-based cyber divisions; India's decentralised, cost-effective forms of

training and forensic analysis; and the United States' advanced jurisprudence and institutional specialisation. By doing so, it emphasised Bangladesh's present framework, which is characterised by new legislative developments such as the Digital Security Act and the Cybersecurity Ordinance 2025 but is hampered by diverse institutions, limited forensic capability, and legal uncertainty.

Key findings suggest that, while Bangladesh has achieved legislative progress, limitations in technical expertise, institutional coordination, and judicial capability remain. According to comparative study, successful digital investigations rely not just on cutting-edge technology, but also on legally applicable norms for the context, appropriately coordinated multi-agency arrangements, and ongoing capacity building. The study thus gives a comprehensive knowledge of how Bangladesh might realistically enhance its cyber investigative environment by learning from these various approaches.

New Insights

The research study emphasises the need of context-dependent technology adoption for cybercrime investigation in Bangladesh, taking into account the country's socioeconomic situation, institutional capability, and legal culture. Yes, sophisticated digital forensic techniques and technologies are required, but their effectiveness is dependent on their successful integration into existing court and policing processes. For example, Colombia's TC2IL exemplifies the benefits of regional specialisation and international collaboration, but replicating similar models in Bangladesh requires overcoming distinct infrastructure and resource restrictions. Similarly, India's decentralised and low-cost innovation initiatives demonstrate that technology deployment does not necessarily have to be expensive, but must prioritise scalability and accessibility in varied geographies.

Another key conclusion is that legal frameworks must evolve with investigating tactics. With ever-evolving technology rewriting the landscape of digital crime on a daily basis, rules must be flexible while precise enough to govern new forms of cybercrime and evidence manipulation. Bangladesh's experience with

the Digital Security Act and the Cybersecurity Ordinance 2025 demonstrates this continuous issue; while these pieces of legislation establish the groundwork, ambiguity in procedural detail and right protection may impede effective enforcement and judicial impartiality. Drawing on the United States' strong jurisprudence where key court cases have defined digital privacy and evidence admissibility Bangladesh must balance legal clarity, procedural fairness, and basic rights while extending investigative capabilities.

The study further emphasises the essential function of capacity building and institutional coordination as a prerequisite for technological and legislative change to yield meaningful results. Sophisticated equipment and legislation are squandered or misused when competent detectives, prosecutors, and judges fail to do their duties. Bangladesh's existing scarcity of cyber forensic specialists and ad hoc inter-agency collaboration highlight flaws that should be remedied by systematic, ongoing professional training and the establishment of a centralised coordinating framework, such as a national CERT-type group.

Practical Recommendations

With these findings in consideration, this study proposes a multi-stage reform strategy for Bangladesh, including short-term, medium-term, and long-term measures.

In the short term, Bangladesh must engage in institutional development and core competency enhancement. This includes improving training for law enforcement and judges on cyber laws and digital forensic techniques, with the help of international agencies like as INTERPOL and organisations like India's NDCRTC and the United States' NCFI. A national cybercrime coordination centre would allow for increased information exchange and standardisation of working processes among police, CID, and Cyber Tribunals. Second, providing precise procedural standards for digital evidence gathering and chain of custody would provide investigators and prosecutors with an instant guarantee.

The mid-term reforms should focus on legislative modifications and infrastructure investment. Updating the Digital Security Act and Cybersecurity Ordinance to remove ambiguous elements, comply with worldwide best practice standards, and strengthen privacy and human rights safeguards will boost the law's credibility and efficacy. Investing in modern digital forensic laboratories with enhanced hardware and software at important regional centres would help to shorten case backlogs and improve evidence reliability. Creating specialised judicial training programs and creating specialist cyber courts staffed by qualified judges will speed up cybercrime trial procedures and increase judgement quality.

Bangladesh needs long-term objectives for the sustainability of systemic institutional development and technical innovation. This means combining cyber investigative capabilities into a comprehensive national cybersecurity framework that includes governmental, business, and academic sectors. Establishing research centres focused on cybercrime and digital forensics may foster innovation specific to the Bangladeshi setting, promoting local solutions and technology. Establishing regional and global cooperation models for collaborative operations and intelligence sharing would improve Bangladesh's capabilities to combat cross-border cybercrime. Finally, raising public awareness and promoting cyber hygiene via education and outreach activities will improve prevention and reporting, complementing law enforcement measures.

Closing Reflection

The digital age brings unique challenges and possibilities for global criminal justice systems. Bangladesh is at a crossroads where technical innovation, developing legal standards, and institutional change might transform its approach to cybercrime. This research underscores the need of striking a balance between efficiency, due process, and human rights protection while conducting digital investigations. To appropriately respond to sophisticated digital offences, the police must deploy technology and intelligence-led policing, while the judiciary must

guarantee that evidence collection and utilisation are carried out in a way that protects constitutional and procedural rights.

This harmonisation necessitates ongoing political will, resource deployment, and cross-sectoral coordination. Bangladesh's continuing legal changes and new institutional efforts are encouraging, but realising their full potential requires smart, incremental reforms that are sensitive to national circumstances and global cybercrime realities. Bangladesh can build a strong digital investigative system that not only solves existing concerns but also forecasts upcoming dangers, resulting in a safer, more equitable digital society.

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Note on conflict of interest

I hereby declare that there is no conflict of interest while preparing this research. I have no financial or personal relationship that may influence the interpretation or publication of the results which are obtained. I declare that I am the sole person who developed, and finalized this work, and I ultimately endorse its publication. I also affirm that I have respected ethical standards and upheld scientific integrity at all times, in line with the standards set by the academic world and those prescribed by this journal.

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